

Quick-response liquid powder display (QR-LPD) is another promising technology targeted at ultra-low-power applications. A few other SED display technologies have emerged in the recent times, but these have not seen much traction in the market."

MEMS-based high-speed mobile displays

Apart from optics and materials, "micro-electromechanical systems (MEMS)-based high-speed mobile displays also form an important category of innovation in recent times," feels Anup Tapadia, founder, TouchMagix Media.

The nature-inspired Mirasol display technology developed by Qualcomm is one example. This display is based on a reflective technology called Interferometric MODulation (IMOD) with MEMS structures at its core. It is extremely low-power yet highly reflective—the display can be seen even in direct sunlight.

The display works by reflecting light so that specific wavelengths interfere with each other to create colour. It was inspired by the phenomenon that makes a butterfly's wings shimmer and is claimed to be the industry's first MEMS display for mobiles.

Last year, Hitachi Display demonstrated a MEMS display prototype at CEATEC 2010. Built on Pixtronic's micro-shutter display technology, this device features very low power consumption—almost half that of LCD counterparts. It measured 6.3 cm (2.5 inches) diagonally, with a QVGA resolution of 320×240 pixels. An important feature of MEMS displays demonstrated by Hitachi is that they can operate in three modes—reflective, transparent and translucent. The transparent mode, for example, can be useful for ubiquitous computing or augmented reality applications.

It is hoped that the technology will be commercialised in the last quarter of this year.

Cool, smart displays

The development of displays that consume less power and emit less heat is an ongoing effort. In fact, this

is considered to be one of the prerequisites for achieving long-term goals of ambient electronics and ubiquitous computing, where displays and computers will merge unobtrusively with our environment.

"Power consumption is a major issue for large displays (greater than 208 cm) and there is a concerted effort by government agencies and industry bodies to introduce regulation that will monitor the power consumed by existing and newer solutions. Definite progress is being made to rein in the power consumption on most of the existing technologies," says Raju.

The use of LEDs and OLEDs is a major trend in terms of low-power efforts. Among the newer technologies in the industry, LPD offers a multiple reduction in power relative to other display technologies for the same brightness. Heating, ventilation and air-conditioning (HVAC) and power drops are the other issues to consider. Power consumed by the video controller chips is also a concern. It can be solved to some extent by adopting the latest FPGA chips that bring in power-efficient design features, such as Xilinx Artix 7. Qualcomm's Mirasol display technology also represents a breakthrough in terms of power efficiency.

Very recently, Samsung Electronics acquired Liguavista—a spin-off of Philips Research Labs which develops ultra-low-power display technologies. Liguavista's electrowetting technology, which operates in transmissive, reflective, transparent and transmissive modes, enables creation of displays with bright, colourful images along with dramatically reduced power consumption. Offering more than twice the transmittance of LCD technology and operating at low frequencies, displays utilising electrowetting are expected to consume just 10 per cent of the battery power of existing display technologies.

Samsung plans to use electrowetting technology in its mobile products and expects great benefits in terms of power consumption and response times. In e-paper applications, for example, the

response time of electrowetting displays will be more than 70 times faster than existing reflective displays, allowing colour video display. If and when Samsung brings out electrowetting-based devices, it will be another landmark in the low-power device market.

Displays that engage, interact

R&D efforts are contributing towards increasing the interactivity of displays—a prerequisite for realising the fantastic dreams of ubiquitous computing. Although real ubiquitous computing is still far away, the industry feels that audience recognition and engagement through interactive displays could be the next major milestone in display technology.

"We believe large displays would no longer be delivering one-way messages in the future. A user will be able to interact with these displays using multi-touch, gestures or even his mobile device," says Tapadia.

A Pune-based company, TouchMagix is evolving as a major player in the interactive market for large-format, interactive, multi-touch displays. The company manufactures a new variety of interactive display solutions like Interactive Floor, Interactive Wall, Touch Table (Interactive Table), Touch Window and 3D Magix-Sense. TouchMagix currently exports products to more than 25 countries and serves clients like Nike, Intel, Reebok, Nokia, Castrol, Times Group, Infosys, Merck and Uniliver.

"I feel we are in for a bright future in irregularly shaped displays. In fact, in our company itself, we have an R&D team working on building interactivity for irregular shaped displays," adds Tapadia.

It seems like the day of ubiquitous computing is not too far off with all these varied improvements in display technology. When the time comes, it will be nothing short of a revolution for it will change the way you interact with the computing systems. ●

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